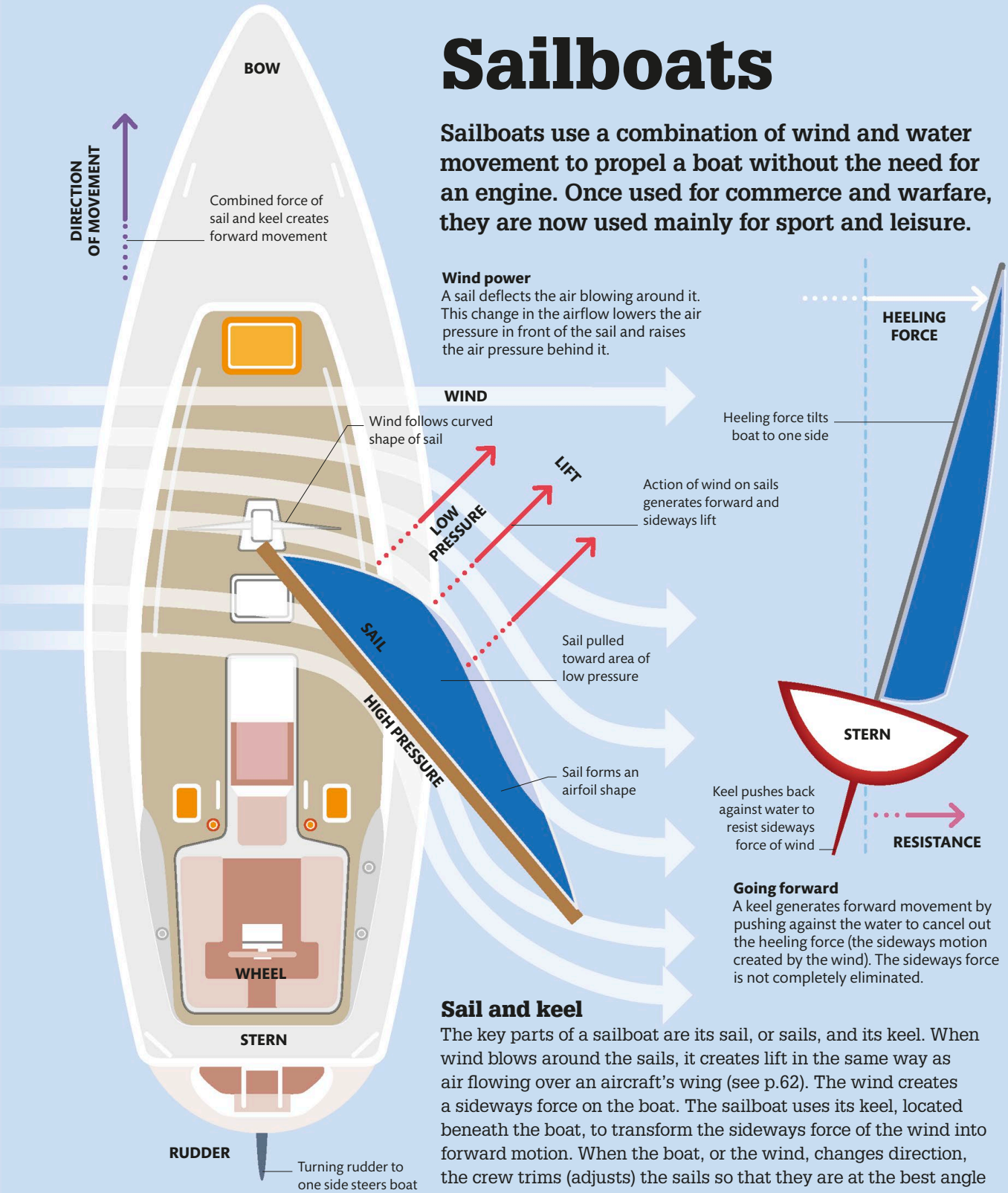
Sailboats

Sailboats use a combination of wind and water movement to propel a boat without the need for an engine. Once used for commerce and warfare, they are now used mainly for sport and leisure.



Wind power
A sail deflects the air blowing around it. This change in the airflow lowers the air pressure in front of the sail and raises the air pressure behind it.

Going forward
A keel generates forward movement by pushing against the water to cancel out the heeling force (the sideways motion created by the wind). The sideways force is not completely eliminated.

Sail and keel

The key parts of a sailboat are its sail, or sails, and its keel. When wind blows around the sails, it creates lift in the same way as air flowing over an aircraft's wing (see p.62). The wind creates a sideways force on the boat. The sailboat uses its keel, located beneath the boat, to transform the sideways force of the wind into forward motion. When the boat, or the wind, changes direction, the crew trims (adjusts) the sails so that they are at the best angle to the wind to generate the most lift.